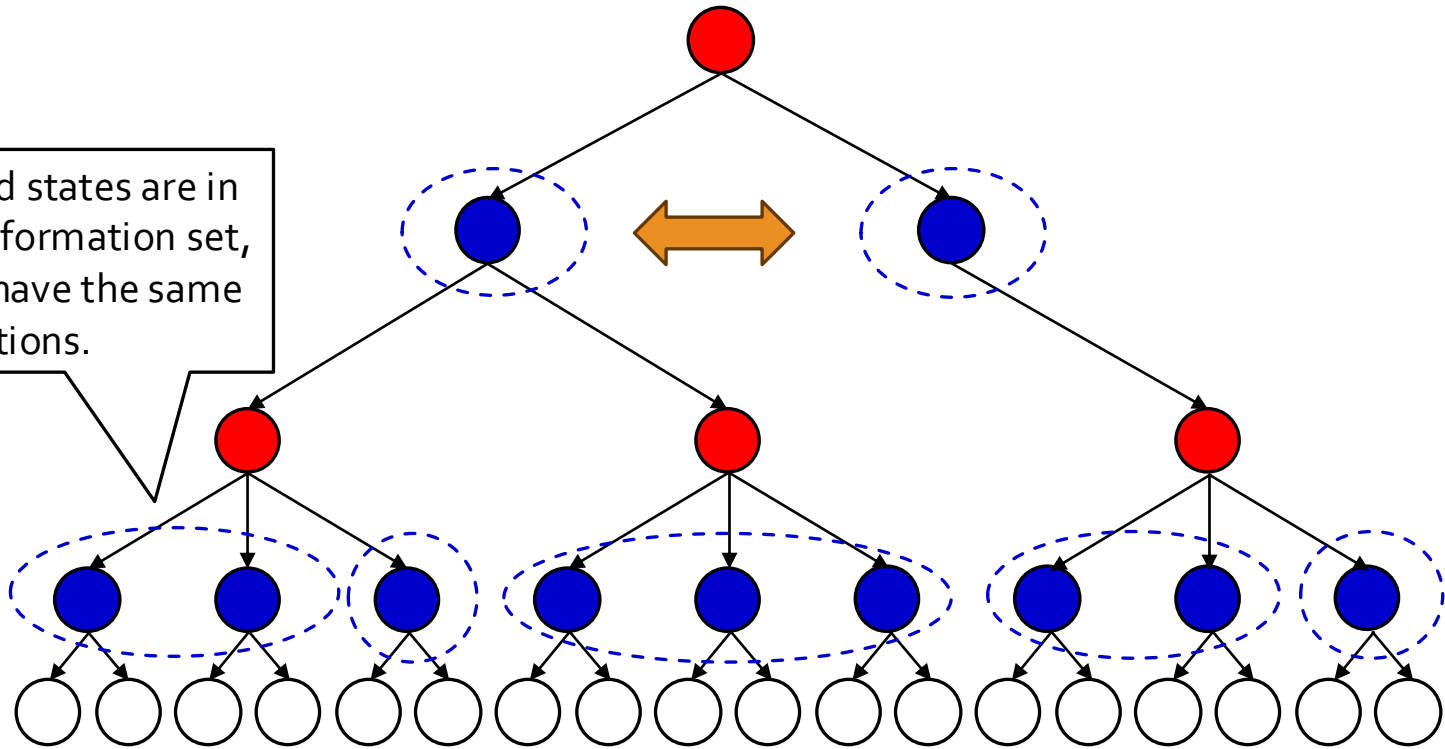




The Value of Information

“You’re not the only one cursed with knowledge” – Thanos, *Avengers: Infinity War*

If two world states are in the same information set, they must have the same actions.



Player i 's knowledge space is some partition of the game states (more generally, a distribution over them).

Recall: a strategy of a player describes their choice in every node that they control.

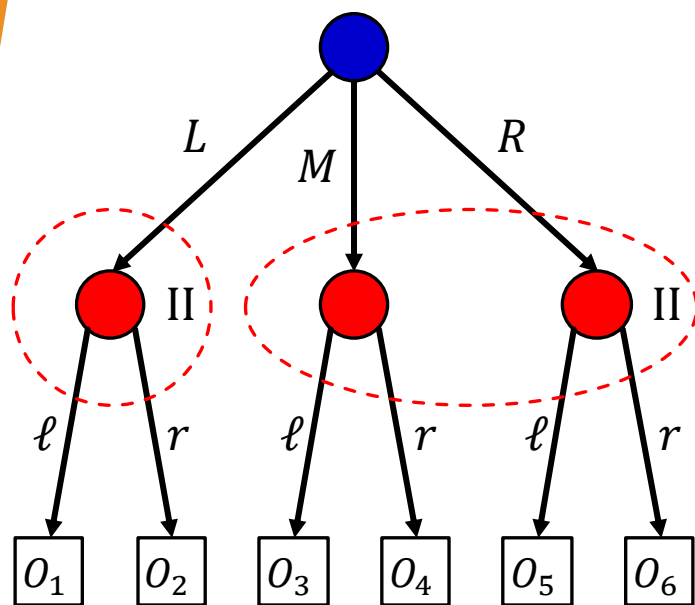
Pure strategy: a deterministic choice of a single strategy.

Mixed strategy: a randomized choice of a strategy.

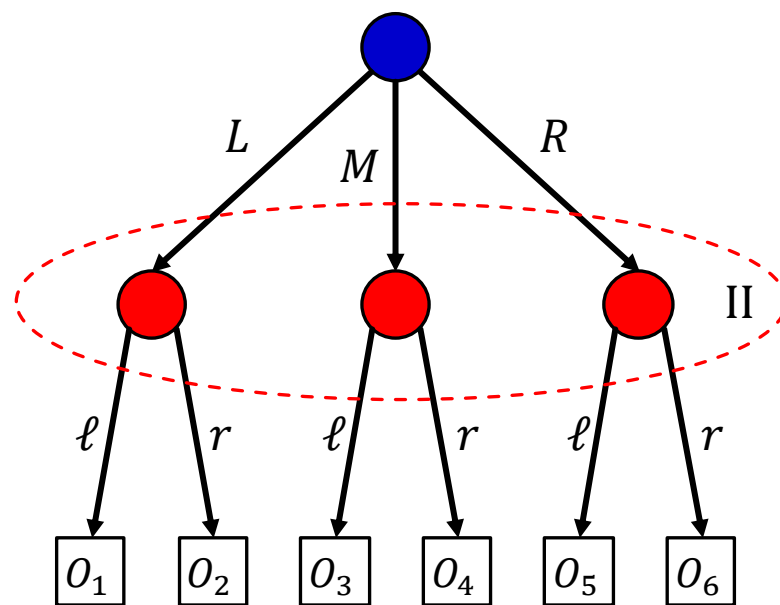
Theorem (Von Neumann's Minimax Theorem): a Nash equilibrium in mixed strategies always exists in zero-sum games.



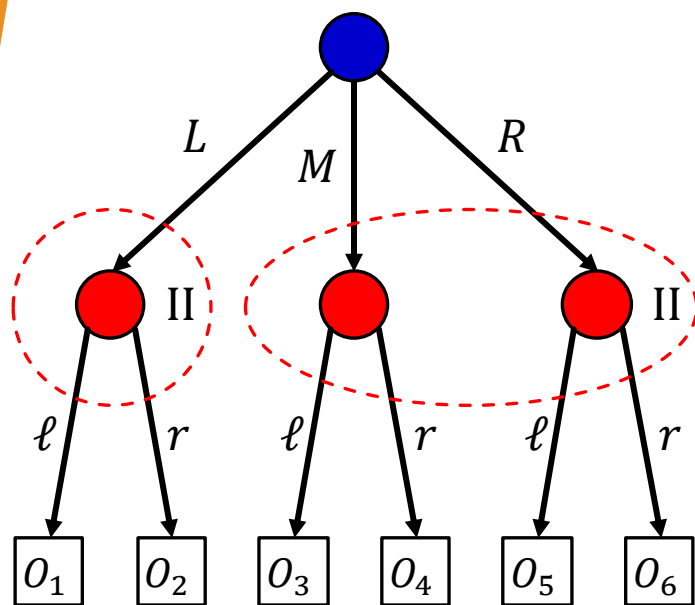
We can think of any 2-player partial information game as a big matrix.



Player II knows whether player I played L , but can't tell M from R .

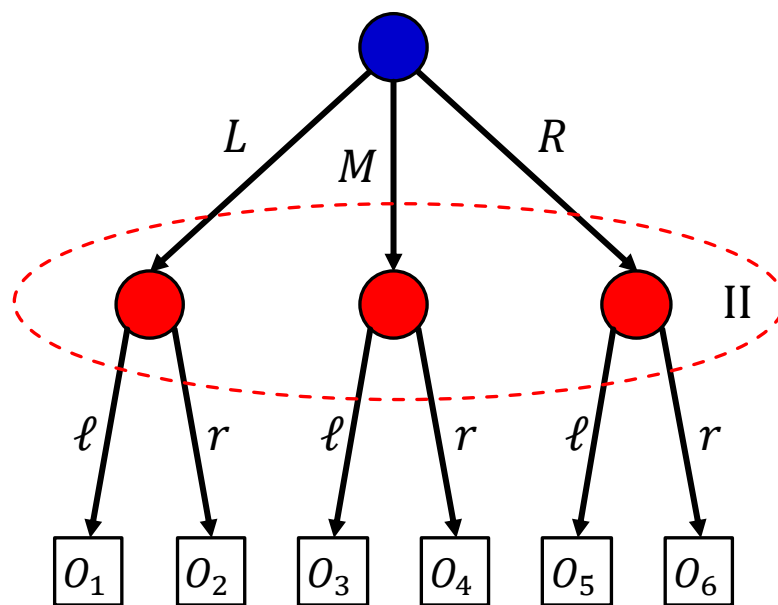


Player II does not know whether player I played L, M or R .



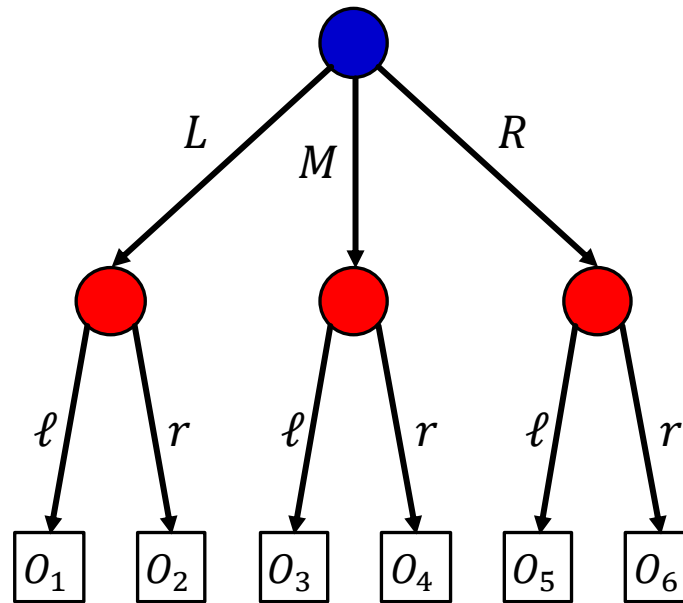
Player II knows whether player I played L , but can't tell M from R .

	$\ell\ell$	ℓr	$r\ell$	rr
L	O_1	O_1	O_2	O_2
M	O_3	O_4	O_3	O_4
R	O_5	O_6	O_5	O_6



Player II does not know whether player I played L, M or R .

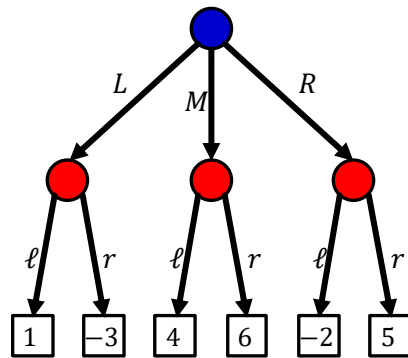
	ℓ	r
L	O_1	O_2
M	O_3	O_4
R	O_5	O_6



Player II knows whether
player I played L , M or R .

	$\ell\ell\ell$	$\ell\ell r$	$\ell r\ell$	$\ell r r$	$r\ell\ell$	$r\ell r$	$rr\ell$	rrr
L	O_1	O_1	O_1	O_1	O_2	O_2	O_2	O_2
M	O_3	O_3	O_4	O_4	O_3	O_3	O_4	O_4
R	O_5	O_6	O_5	O_6	O_5	O_6	O_5	O_6

More Information $\Rightarrow$ more strategies



	$\ell\ell\ell$	$\ell\ell r$	$\ell r\ell$	$\ell r r$	$r\ell\ell$	$r\ell r$	$rr\ell$	rrr
L	1	1	1	1	-3	-3	-3	-3
M	4	4	6	6	4	4	6	6
R	-2	5	-2	5	-2	5	-2	5

We can describe a zero-sum game by the payments that player 1 receives at each terminal node.

Obtain a single matrix:

- Row indices correspond to pure strategies of player 1.
- Column indices correspond to pure strategies of player 2.



	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
a_1								
a_2								
a_3								

We refer to the payoff matrix as A .

A mixed strategy profile $(\vec{p}, \vec{q})$ is a distribution over players' pure strategies.

		q_1	q_2	q_3	q_4	q_5	q_6	q_7	q_8
		b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
p_1	a_1								
p_2	a_2								
p_3	a_3								

We refer to the payoff matrix as A .

A mixed strategy profile $(\vec{p}, \vec{q})$ is a distribution over players' pure strategies.

Player 1's utility is $\vec{p}A\vec{q}^T$

Player 2's utility is $-\vec{p}A\vec{q}^T$

		q_1	q_2	q_3	q_4	q_5	q_6	q_7	q_8
		b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
p_1	a_1								
p_2	a_2								
p_3	a_3								

We refer to the payoff matrix as A .

A mixed strategy profile $(\vec{p}, \vec{q})$ is a distribution over players' pure strategies.

The maximin value:

$$v_+ = \max_{\vec{p} \in \Delta(S_1)} \min_{\vec{q} \in \Delta(S_2)} \vec{p} A \vec{q}^T$$

The minimax value:

$$v_- = \min_{\vec{q} \in \Delta(S_2)} \max_{\vec{p} \in \Delta(S_1)} \vec{p} A \vec{q}^T$$


$$v_+ = \max_{\vec{p} \in \Delta(S_1)} \min_{\vec{q} \in \Delta(S_2)} \vec{p} A \vec{q}^T$$

Player 1 tells Player 2 their strategy, and then Player 2 picks a strategy $\vec{q}$ that minimizes the utility of Player 1.

Thus Player 1 wants to pick a strategy $\vec{p}^*$ that will maximize their payoff.

If Player 1 plays $\vec{p}^*$, they are guaranteed to be paid at least v_+ no matter what Player 2 plays.

The Minimax Theorem

Von Neumann's Minimax Theorem:
maximin equals minimax: $v_+ = v_-$

Von Neumann's Minimax Theorem: maximin equals minimax: $v_+ = v_-$

Suppose that player 1 switches to playing $\vec{p}$ instead of $\vec{p}^*$.

The strategy $\vec{q}^*$ satisfies for any $\vec{p} \in \Delta(S_1)$

$$-u_1(\vec{p}, \vec{q}^*) = u_2(\vec{p}, \vec{q}^*) \geq -v_- = -v_+ = -u_1(\vec{p}^*, \vec{q}^*)$$

Thus, for any strategy $\vec{p} \in \Delta(S_1)$:

$$u_1(\vec{p}^*, \vec{q}^*) \geq u_1(\vec{p}, \vec{q}^*)$$

In other words, $\vec{p}^*$ is a best response to $\vec{q}^*$.

Von Neumann's Minimax Theorem: maximin equals minimax: $v_+ = v_-$

Suppose that player 2 switches to playing $\vec{q}$ instead of $\vec{q}^*$.

The strategy $\vec{p}^*$ satisfies for any $\vec{q} \in \Delta(S_2)$

$$-u_2(\vec{p}^*, \vec{q}) = u_1(\vec{p}^*, \vec{q}) \geq v_+ = v_- = -u_2(\vec{p}^*, \vec{q}^*)$$

Thus, for any strategy $\vec{q} \in \Delta(S_2)$:

$$u_2(\vec{p}^*, \vec{q}^*) \geq u_2(\vec{p}^*, \vec{q})$$

In other words, $\vec{q}^*$ is a best response to $\vec{p}^*$.

If player 1 plays their maximin strategy and player 2 plays their minimax strategy, the resulting strategy profile is a Nash equilibrium.

Proposition:

For any strategy $\vec{p}$ by player 1, there exists a pure strategy best response by player 2.

Proof:

Let $\vec{q}^*$ be a best response to $\vec{p}$ and let $\ell \in \arg \min_{j \in \text{supp}(\vec{q}^*)} \sum_i a_{ij} p_i$. Then:

$$\begin{aligned} \sum_j \sum_i a_{ij} p_i q_j^* &= \sum_j q_j^* \left(\sum_i a_{ij} p_i \right) \\ &\geq \sum_j q_j^* \left(\sum_i a_{i\ell} p_i \right) \\ &= \left(\sum_i a_{i\ell} p_i \right) \left(\sum_j q_j^* \right) = \sum_i a_{i\ell} p_i \end{aligned}$$

Proposition:

For any strategy $\vec{p}$ by player 1, there exists a pure strategy best response by player 2.

In particular: maximin/minimax values can be computed by taking the max over players' pure strategies.

The maximin value:

$$v_+ = \max_{s_1 \in S_1} \min_{\vec{q} \in \Delta(S_2)} A_{s_1} \vec{q}^T$$

The minimax value:

$$v_- = \min_{s_2 \in S_2} \max_{\vec{p} \in \Delta(S_1)} \vec{p} A^{s_2}$$

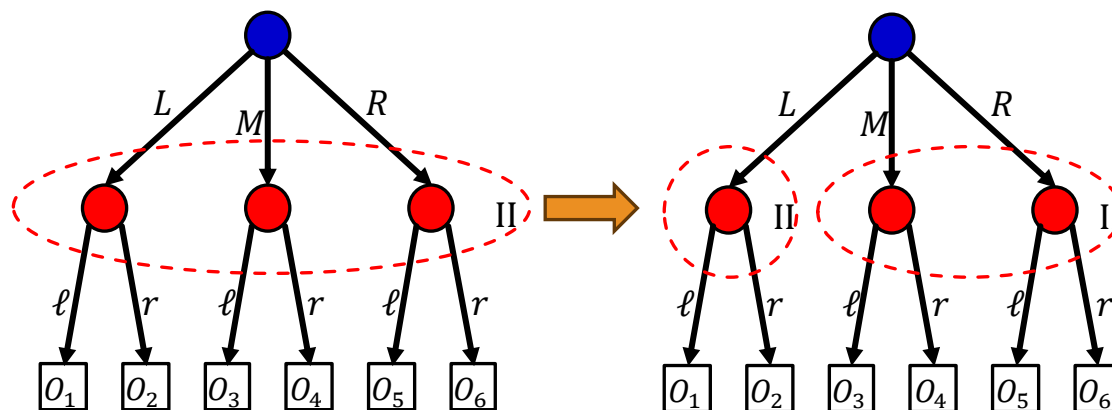
Where A_{s_1} is the s_1 -th row of the matrix A , and A^{s_2} is the s_2 -th column of the matrix A .

We are given a 2-player game in extensive form.

Each player i has a partition of the state space into information sets U_i

Suppose that we **refine** player i 's information sets: $U'_i \subseteq U_i$.

In other words, U'_i is a finer partition of player i 's information, splitting some of their information sets.



Theorem: let Γ' be the game induced from the game Γ by refining the information sets of player i , i.e., $U'_i \subseteq U_i$.

All other players' information sets are the same.

Then player i 's utility in the resulting Nash equilibrium is weakly higher.

Proof: let S_j be the set of pure strategies of player j in the zero-sum game.

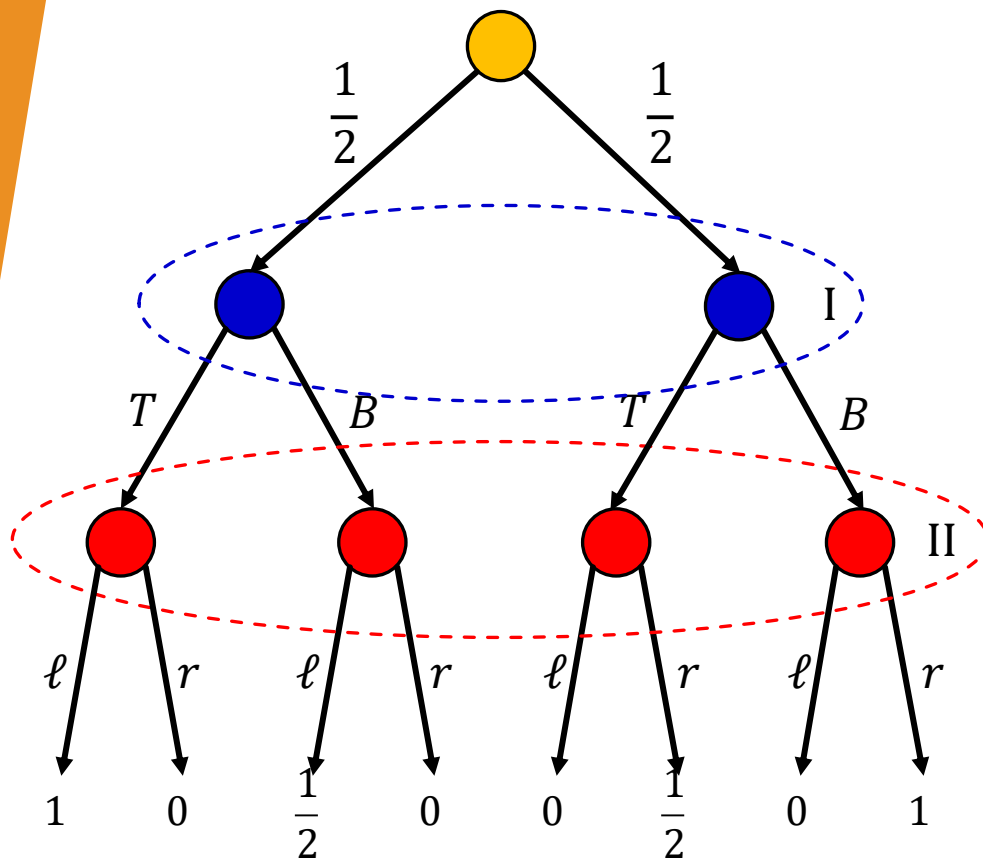
Assume that we refined player 1's information space, their pure strategy space is greater: $S_1 \subseteq S'_1$. For player 2 we have $S_2 = S'_2$.

Thus:

$$\max_{s_1 \in S_1} \min_{\vec{q} \in \Delta(S_2)} A_{s_1} \vec{q}^T \leq \max_{s_1 \in S'_1} \min_{\vec{q} \in \Delta(S_2)} A_{s_1} \vec{q}^T$$



If the game is not zero-sum, then knowing more may **help** or **hurt**.

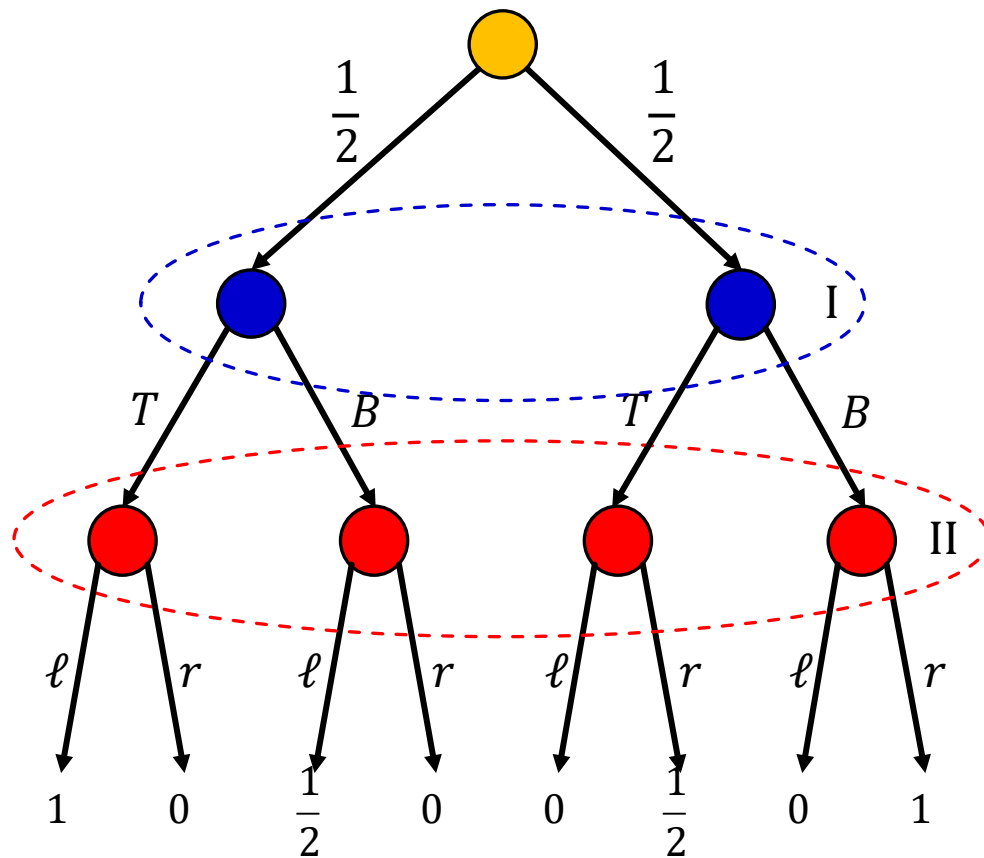


Zero-Sum game: payments to player 1 are listed.

One of the subtrees is selected at random.

Players don't know which one.

Player 2 does not know whether player 1 picked T or B .



	ℓ	r
T	1	0
B	$\frac{1}{2}$	0

	ℓ	r
T	0	$\frac{1}{2}$
B	0	1



$$\frac{1}{2} \times_B^T \begin{array}{c|c} \ell & r \\ \hline 1 & 0 \\ \hline \frac{1}{2} & 0 \end{array} + \frac{1}{2} \times_B^T \begin{array}{c|c} \ell & r \\ \hline 0 & \frac{1}{2} \\ \hline 0 & 1 \end{array} =$$

$$\begin{array}{c} T \\ B \end{array} \begin{array}{c|c} \ell & r \\ \hline \frac{1}{2} & \frac{1}{4} \\ \hline \frac{1}{4} & \frac{1}{2} \end{array}$$

	ℓ	r
T	$\frac{1}{2}$	$\frac{1}{4}$
B	$\frac{1}{4}$	$\frac{1}{2}$

$$u_1(T, q) = \frac{1}{2}q + \frac{1}{4}(1 - q) = \frac{1}{4} + \frac{1}{4}q$$

$$u_1(B, q) = \frac{1}{4}q + \frac{1}{2}(1 - q) = \frac{1}{2} - \frac{1}{4}q$$

$$u_1(T, q) \geq u_1(B, q) \Leftrightarrow \frac{1}{4} + \frac{1}{4}q \geq \frac{1}{2} - \frac{1}{4}q \Leftrightarrow \frac{1}{2}q \geq \frac{1}{4} \Leftrightarrow q \geq \frac{1}{2}$$

$$BR_1(q) = \begin{cases} T & \text{if } q > \frac{1}{2} \\ B & \text{if } q < \frac{1}{2} \\ \text{anything} & \text{if } q = \frac{1}{2} \end{cases}$$

	ℓ	r
T	$\frac{1}{2}, \frac{1}{2}$	$\frac{1}{4}, \frac{1}{4}$
B	$\frac{1}{4}, \frac{1}{4}$	$\frac{1}{2}, \frac{1}{2}$

$$u_2(p, \ell) = \left(-\frac{1}{2}\right) \cdot p + \left(-\frac{1}{4}\right) \cdot (1 - p) = -\left(\frac{1}{4} + \frac{1}{4}p\right)$$

$$u_2(p, r) = \left(-\frac{1}{4}\right) \cdot p + \left(-\frac{1}{2}\right) \cdot (1 - p) = -\frac{1}{2} + \frac{1}{4}p$$

$$u_2(p, \ell) \geq u_2(p, r) \Leftrightarrow -\frac{1}{4} - \frac{1}{4}p \geq -\frac{1}{2} + \frac{1}{4}p \Leftrightarrow \frac{1}{2}p \leq \frac{1}{4} \Leftrightarrow p \leq \frac{1}{2}$$

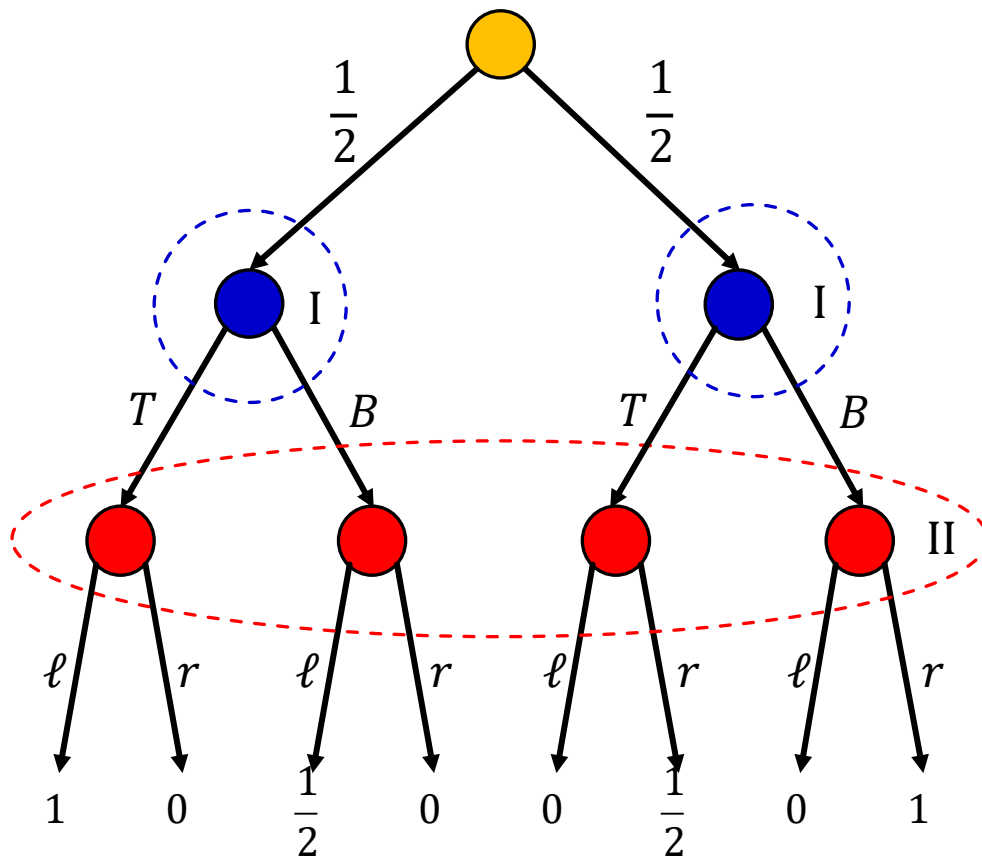
$$BR_2(p) = \begin{cases} \ell & \text{if } p < \frac{1}{2} \\ r & \text{if } p > \frac{1}{2} \\ \text{anything} & \text{if } p = \frac{1}{2} \end{cases}$$

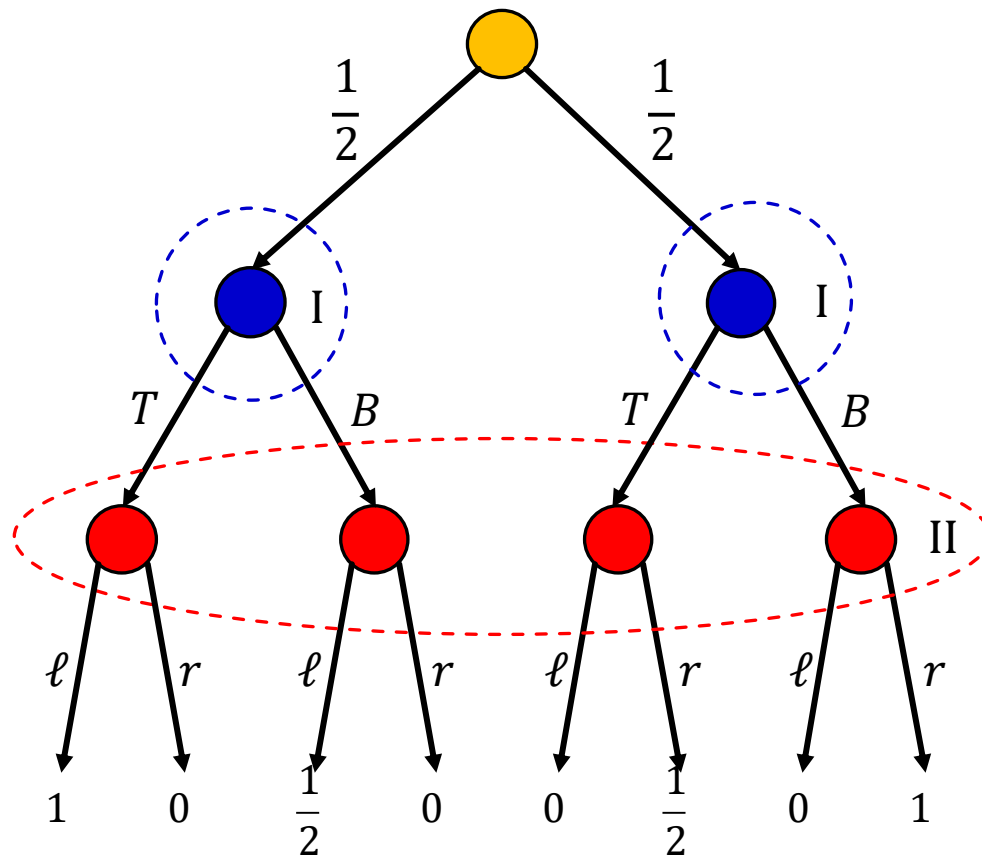
	ℓ	r
T	$\frac{1}{2}, \frac{1}{2}$	$\frac{1}{4}, \frac{1}{4}$
B	$\frac{1}{4}, \frac{1}{4}$	$\frac{1}{2}, \frac{1}{2}$

Thus, the Nash equilibrium of this game is

$$\left(\frac{1}{2}T + \frac{1}{2}B, \frac{1}{2}\ell + \frac{1}{2}r \right)$$

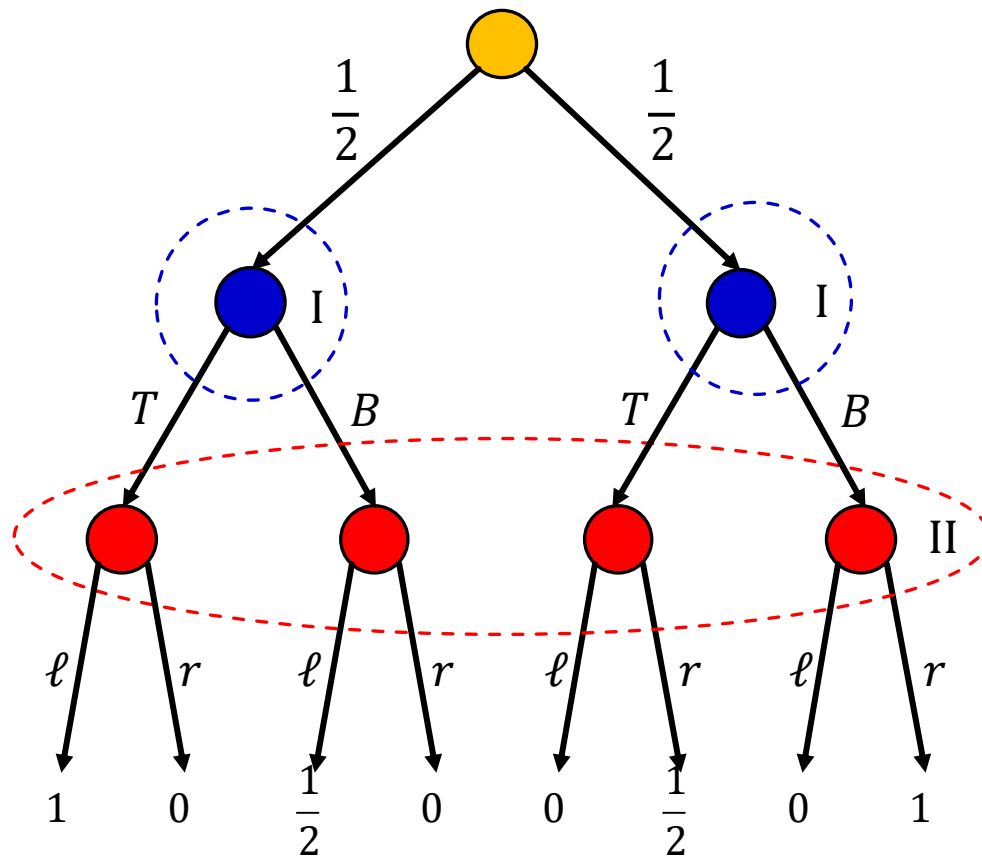
$$BR_1(q) = \begin{cases} T & \text{if } q > \frac{1}{2} \\ B & \text{if } q < \frac{1}{2} \\ \text{anything} & \text{if } q = \frac{1}{2} \end{cases}, \quad BR_2(p) = \begin{cases} \ell & \text{if } p < \frac{1}{2} \\ r & \text{if } p > \frac{1}{2} \\ \text{anything} & \text{if } p = \frac{1}{2} \end{cases}$$





	ℓ	r
T	1	0
B	$\frac{1}{2}$	0

	ℓ	r
T	0	$\frac{1}{2}$
B	0	1



	ℓ	r
T	1	0
B	$\frac{1}{2}$	0

	ℓ	r
T	0	$\frac{1}{2}$
B	0	1

	ℓ	r
T	1	0
B	$\frac{1}{2}$	0

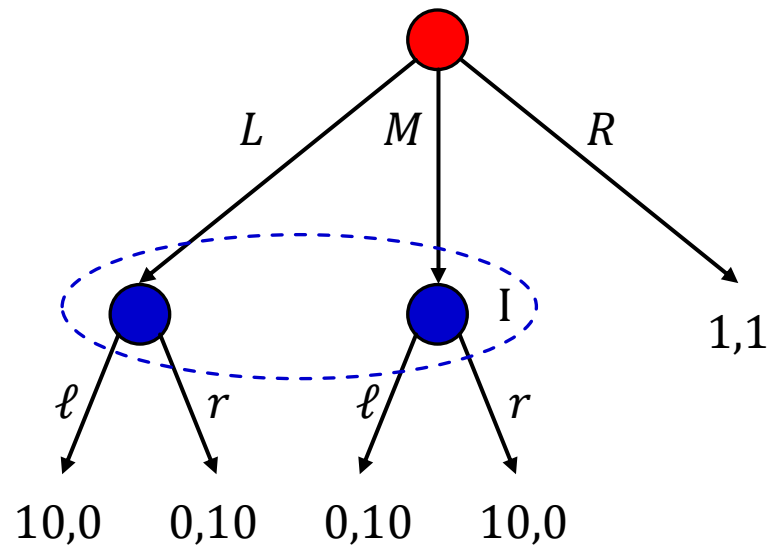
	ℓ	r
T	0	$\frac{1}{2}$
B	0	1

Player 2 does not know whether player 1 played T or B , nor do they know what nature chose.

Player 2's utility from any strategy $(q, 1 - q)$ is

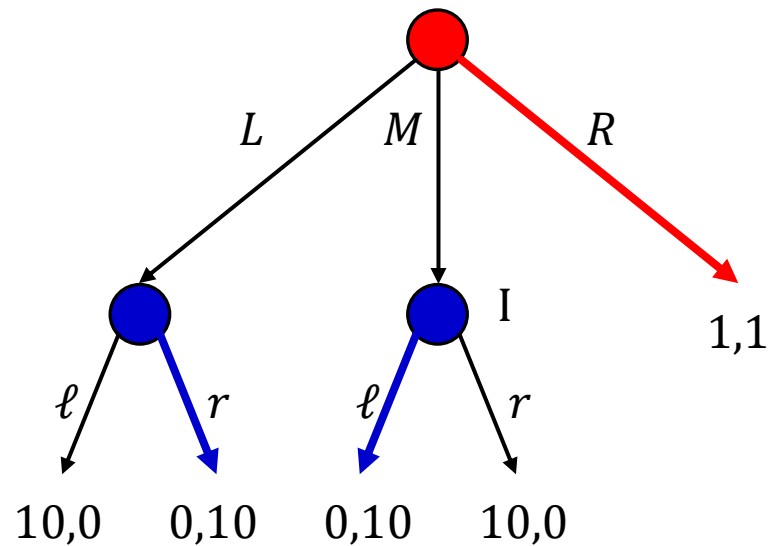
$$\begin{aligned} & \frac{1}{2} \times ((-1) \cdot q + 0 \cdot (1 - q)) + \frac{1}{2} (0 \cdot q + (-1) \cdot (1 - q)) \\ &= -\frac{1}{2}q - \frac{1}{2}(1 - q) = -\frac{1}{2} \end{aligned}$$

(T, B) is also a best response against $(q, 1 - q)$ for any q . Their expected utility is $\frac{1}{2}$. This is better than the $\frac{3}{8}$ utility they receive when they have less information.



Player 1 can choose R , in which case both players are paid 1. Player 2 cannot tell whether Player 1 chose L or M .

The only Nash equilibrium in this case is $\left(\frac{1}{2}L + \frac{1}{2}M, \frac{1}{2}\ell + \frac{1}{2}r\right)$, with both players receiving an expected utility of 5.

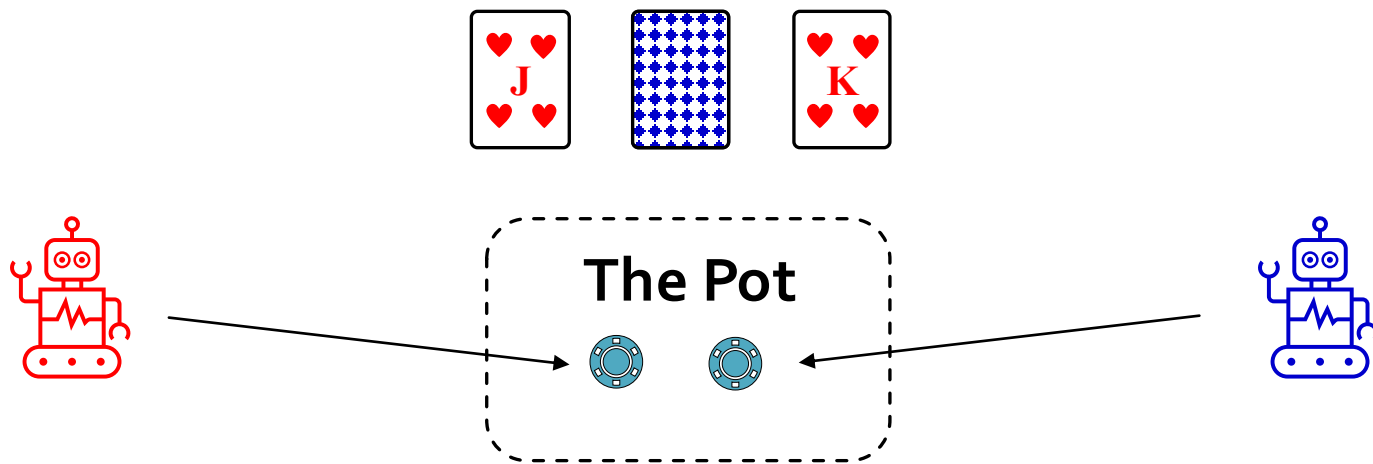


Suppose that Player 2 can tell whether Player 1 chose L or M . The only Nash equilibrium here is highlighted above.

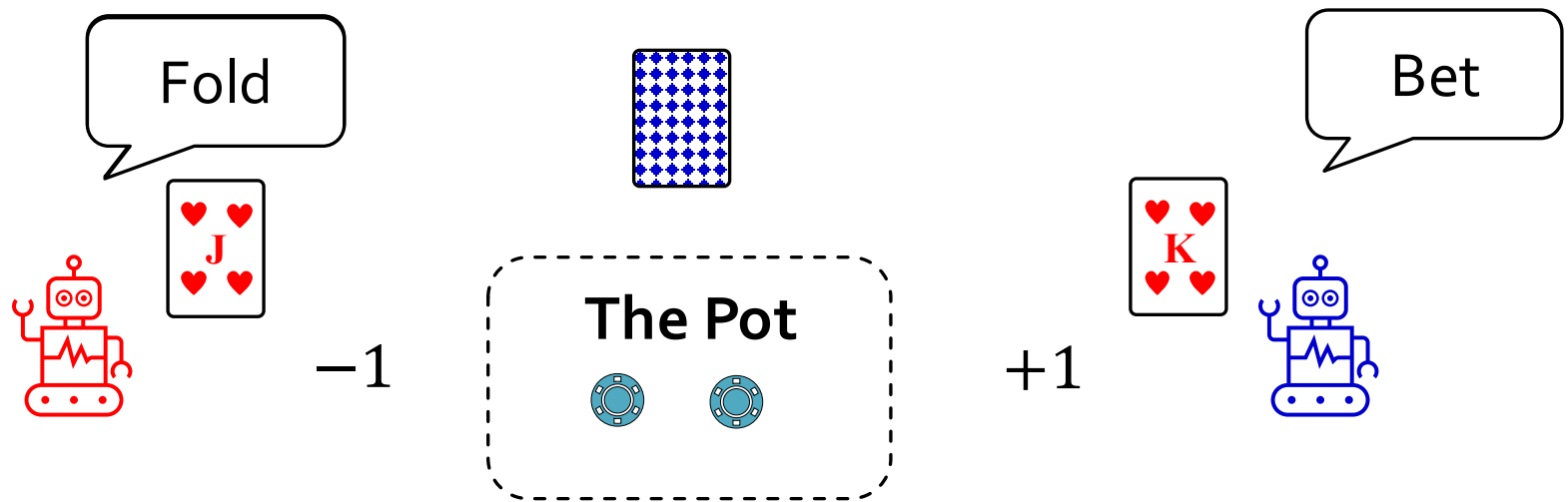
Under L and M , Player 2 picks the action that offers them a payoff of 10 and a payoff of 0 to player 1. Thus, player 1 prefers to play R and get a payoff of 0.



Example: Kuhn's Poker

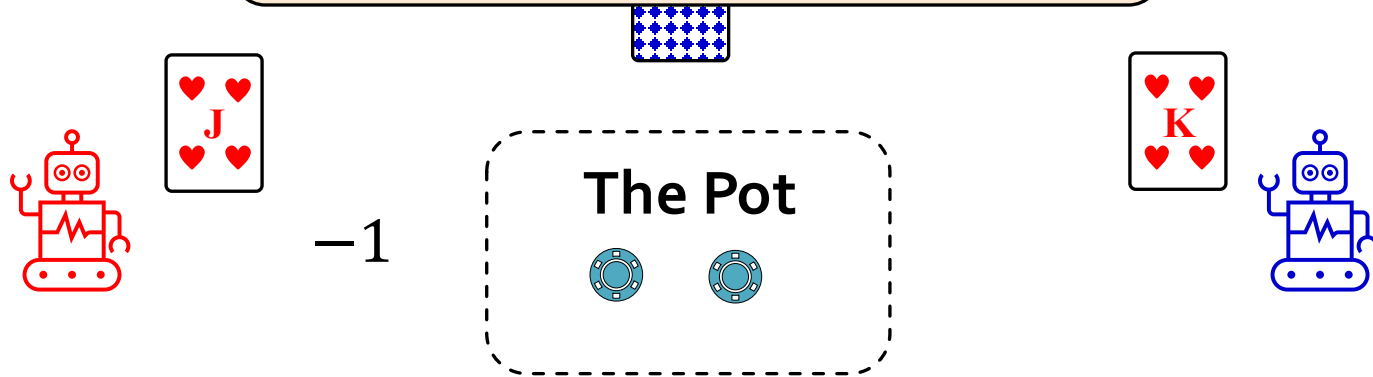


- Each player puts one chip in the pot.
- Each player receives one card, face down.
- The remaining card is set aside, unseen by either player.
- Every possible card assignment is equally likely (probability $\frac{1}{6}$).

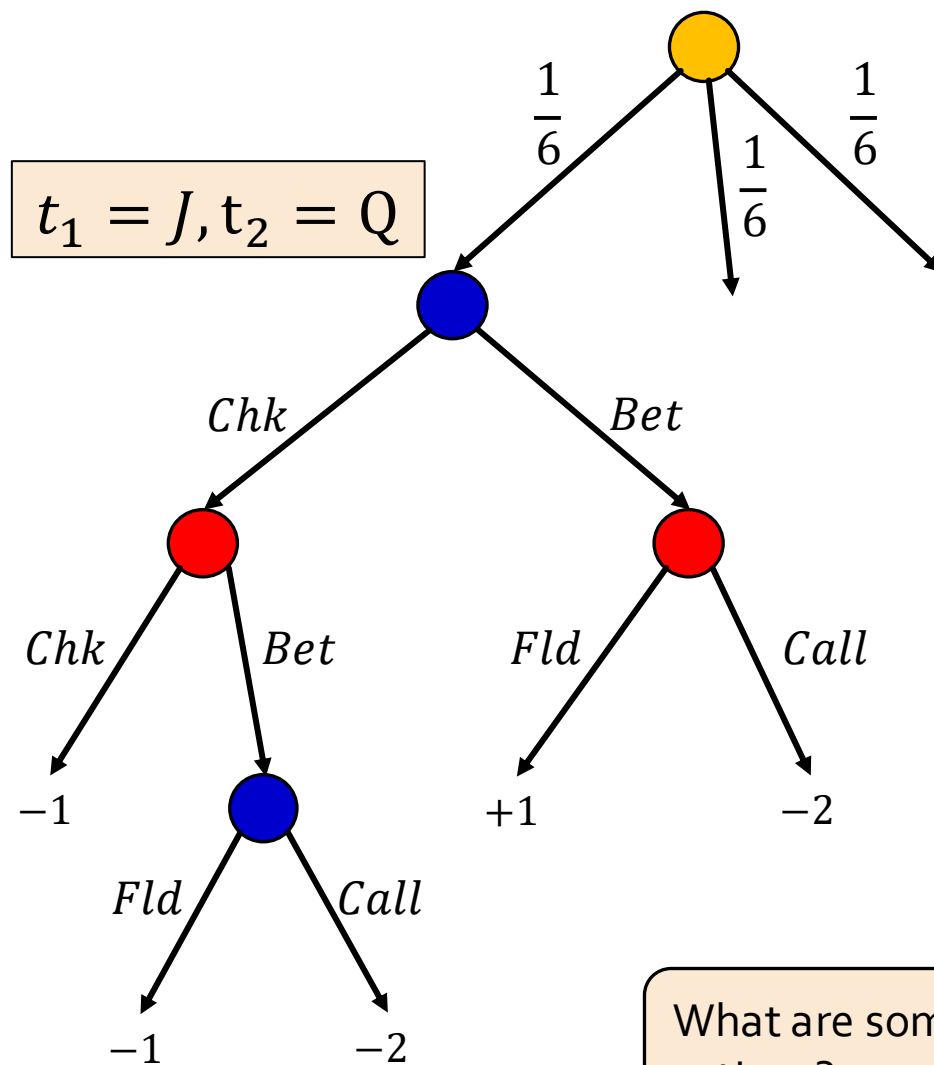


- Player 1 chooses whether to **Check** or to **Bet**. Player 2 chooses an action next.
- If player 1 checks, then player 2 either **checks** or **bets**.
 - If **both players check**, the higher card wins (+1 for the winner, -1 for the loser).
 - If player 2 bets, then player 1 either **folds** or **calls**.
 - If player 1 folds then player 2 wins the pot (-1 for player 1, $+1$ for player 2).
 - If player 1 calls, the higher card wins the pot ($+2$ for the winner, -2 for the loser).
- If player 1 bets, then player 2 either **folds** or **calls**.
 - If player 2 folds, then player 1 wins the pot ($+1$ for player 1, -1 for player 2).
 - If player 2 calls, then the higher card wins the pot ($+2$ for the winner, -2 for the loser).

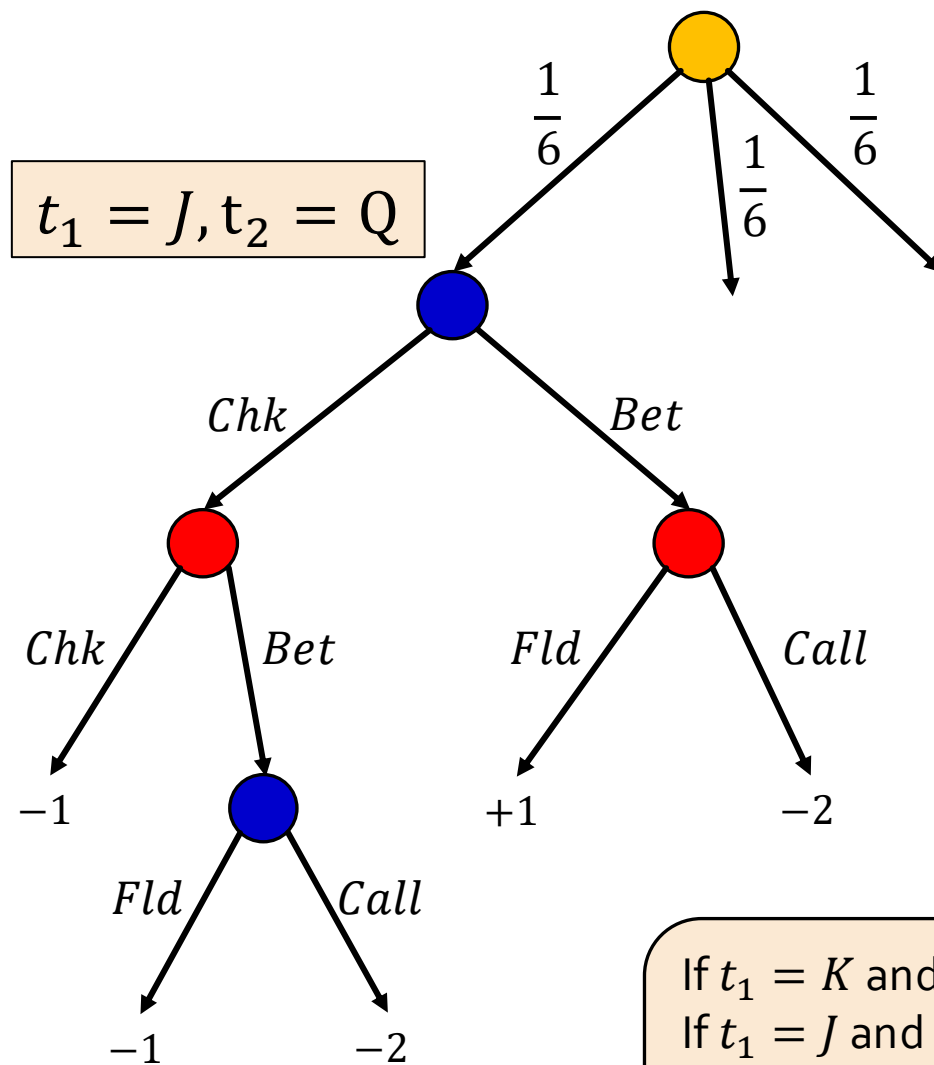
1. Draw the game tree (at least one subtree after the nature move).
2. What are players' information sets?



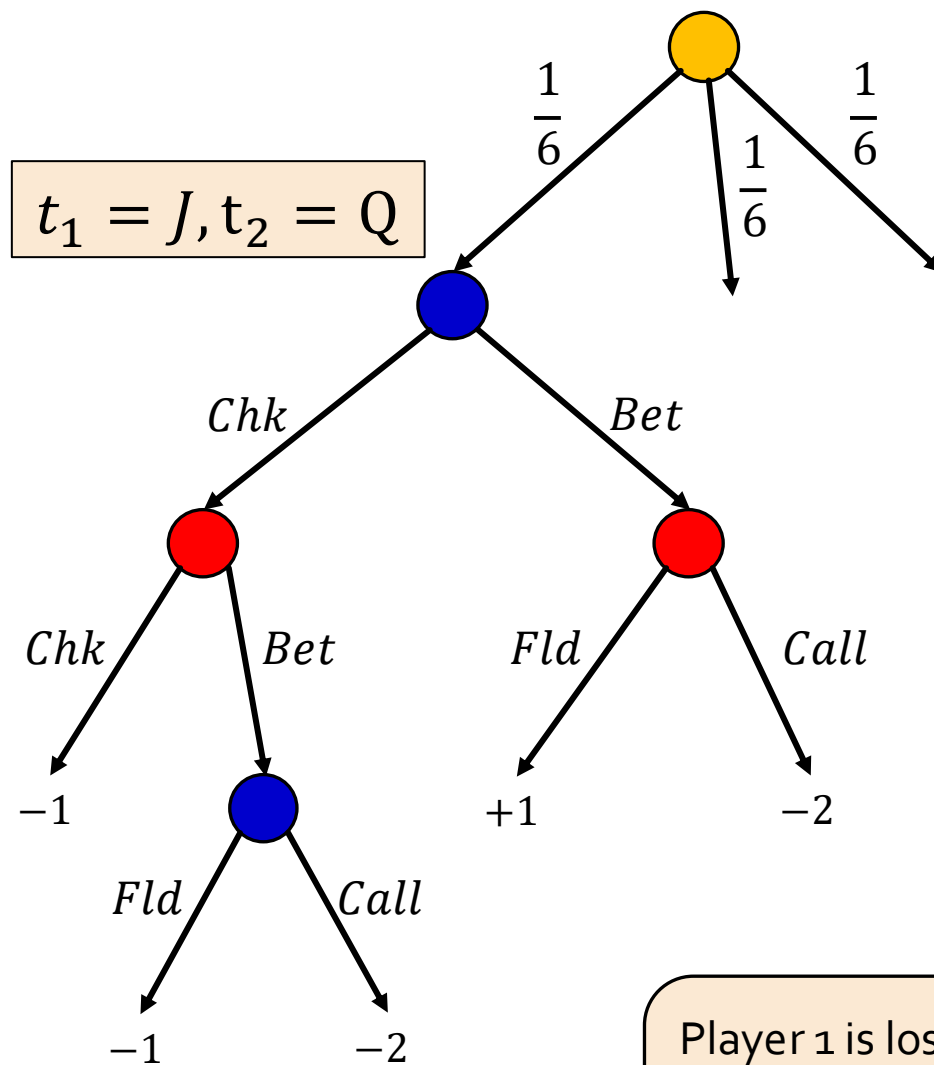
- Player 1 chooses whether to **Check** or to **Bet**. Player 2 chooses an action next.
- If player 1 checks, then player 2 either **checks** or **bets**.
 - If **both players check**, the higher card wins (+1 for the winner, -1 for the loser).
 - If player 2 bets, then player 1 either **folds** or **calls**.
 - If player 1 folds then player 2 wins the pot (-1 for player 1, +1 for player 2).
 - If player 1 calls, the higher card wins the pot (+2 for the winner, -2 for the loser).
- If player 1 bets, then player 2 either **folds** or **calls**.
 - If player 2 folds, then player 1 wins the pot (+1 for player 1, -1 for player 2).
 - If player 2 calls, then the higher card wins the pot (+2 for the winner, -2 for the loser).



What are some immediate equilibrium actions?

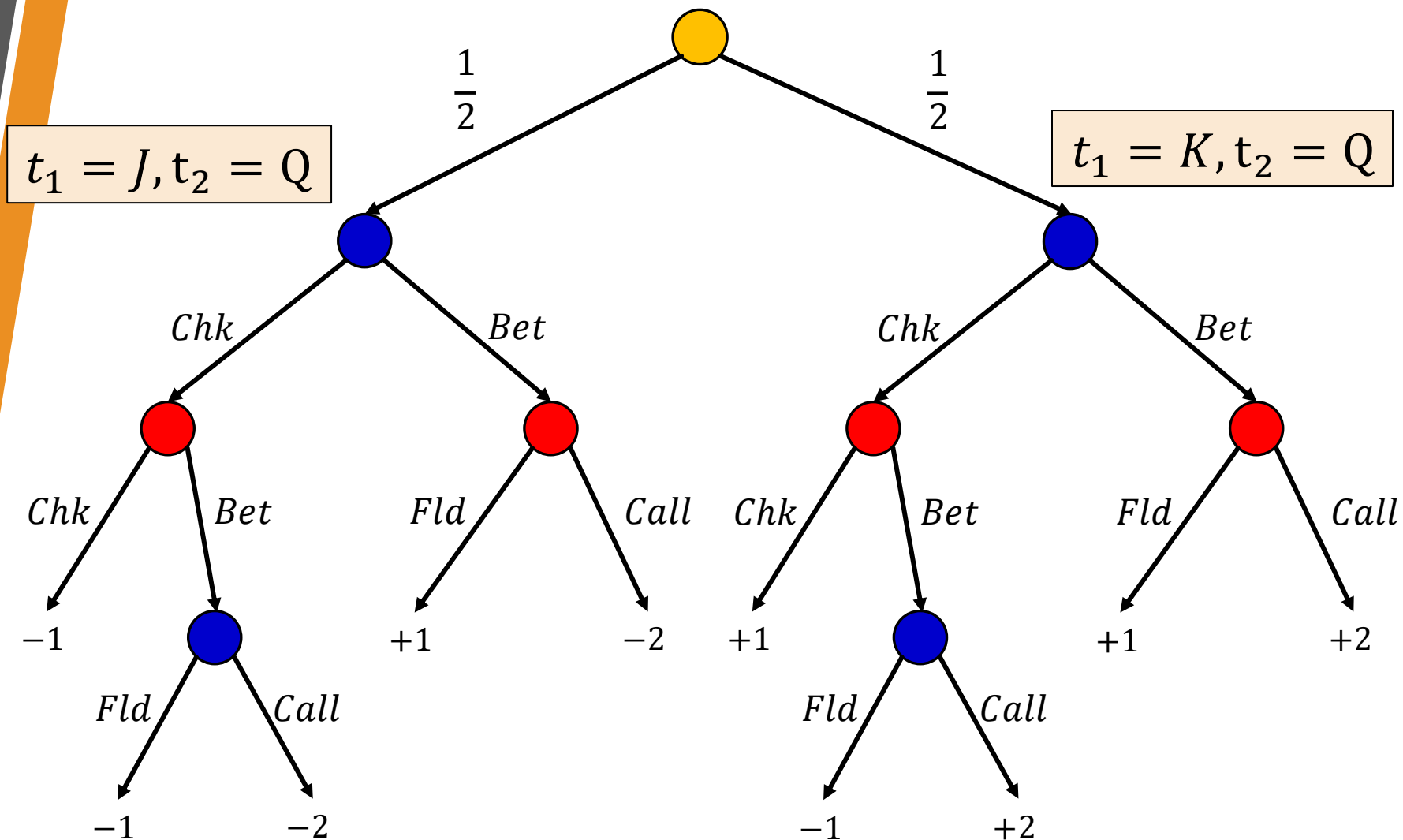


If $t_1 = K$ and $(Chk, Bet) \Rightarrow P1$ Call
 If $t_1 = J$ and $(Chk, Bet) \Rightarrow P1$ Fld
 If $t_2 = K$ and $P1$ Bet $\Rightarrow P2$ Call
 If $t_2 = K$ and $P1$ Chk $\Rightarrow P2$ Bet
 If $t_2 = J$ and $P1$ Bet $\Rightarrow P2$ Fld



Player 1 is losing this hand, and they know it.

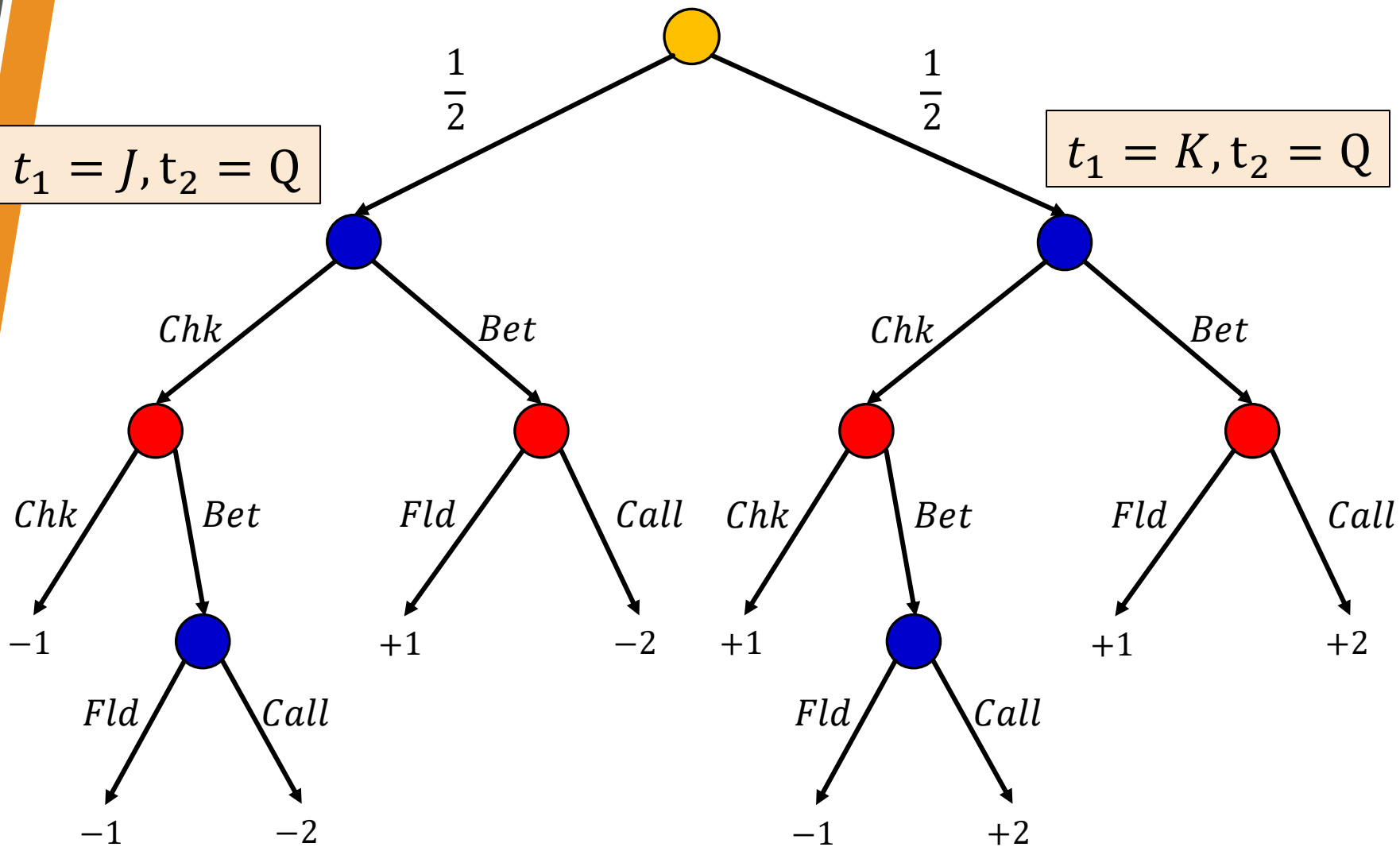
However... Player 2 does not know this.



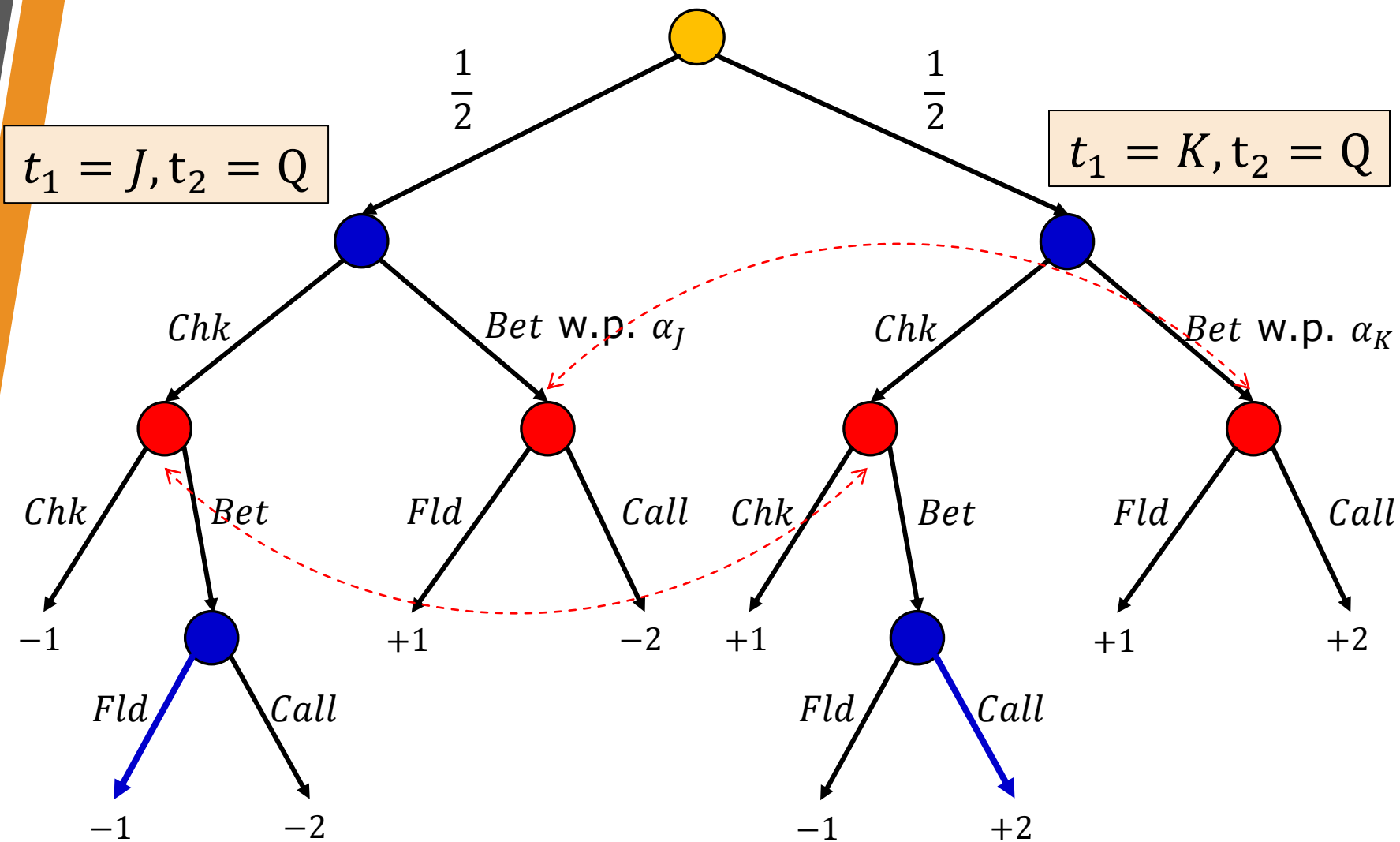
From Player 2's perspective

$t_1 = J, t_2 = Q$

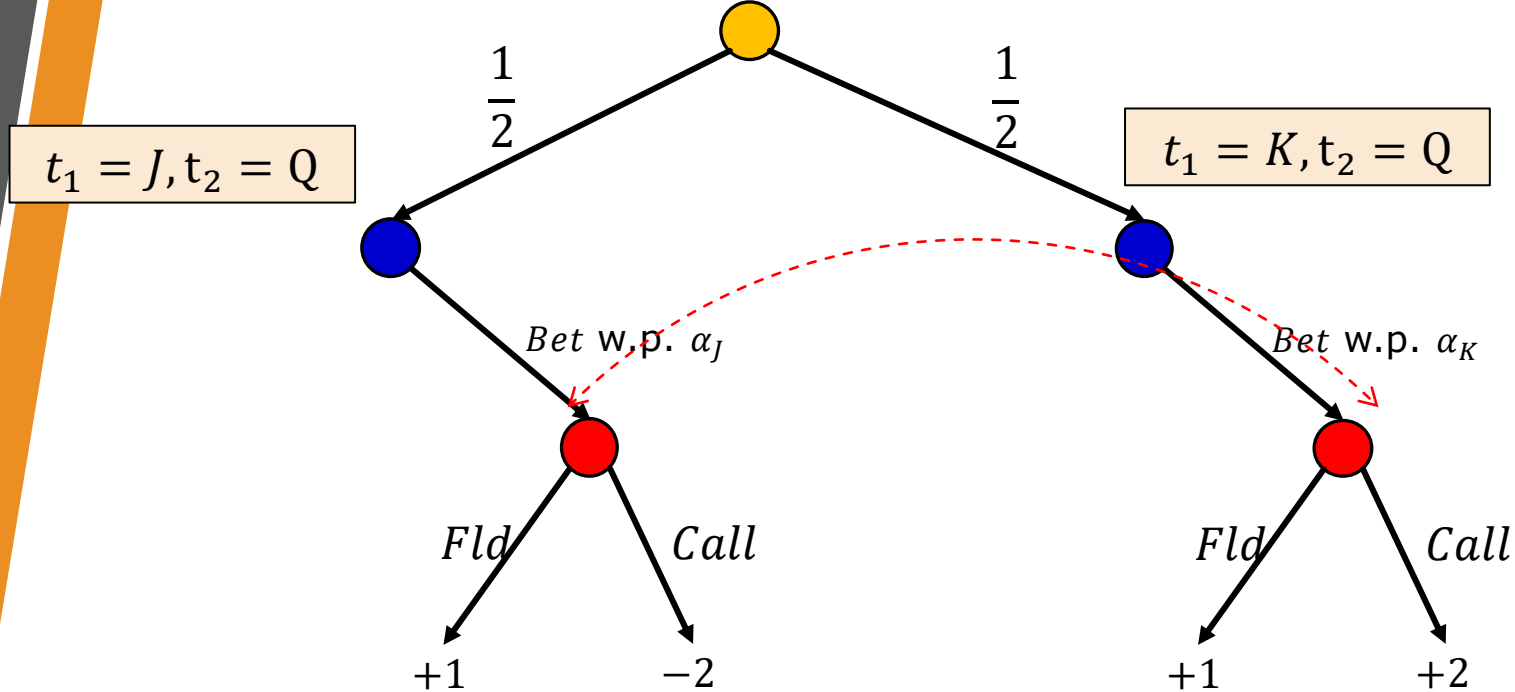
$t_1 = K, t_2 = Q$



What is the equilibrium strategy given $t_1 = J, t_2 = Q$?



Let δ_Q be P2's call probability given that P1 bet.



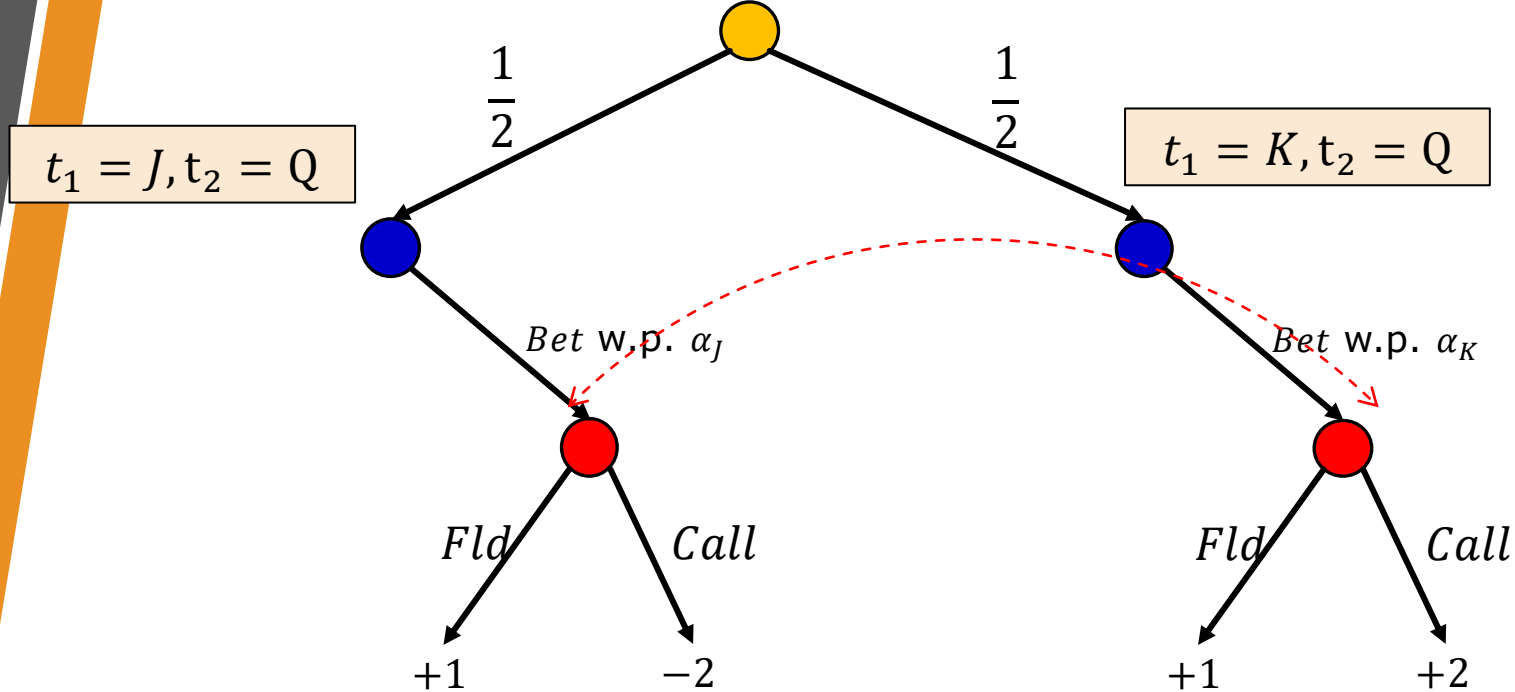
P2's utility from calling is $\Pr[t_1 = J \mid P1 \text{ bet}] \cdot 2 + \Pr[t_1 = K \mid P1 \text{ bet}] \cdot (-2)$

$$\Pr[t_1 = J \mid P1 \text{ bet}] = \Pr[P1 \text{ bet} \mid t_1 = J] \cdot \frac{\Pr[t_1 = J]}{\Pr[P1 \text{ bet}]}$$

$$\Pr[P1 \text{ bet}] = \Pr[P1 \text{ bet} \mid t_1 = K] \cdot \Pr[t_1 = K] + \Pr[P1 \text{ bet} \mid t_1 = J] \cdot \Pr[t_1 = J] = \frac{1}{2}(\alpha_K + \alpha_J)$$

Thus P2's utility is

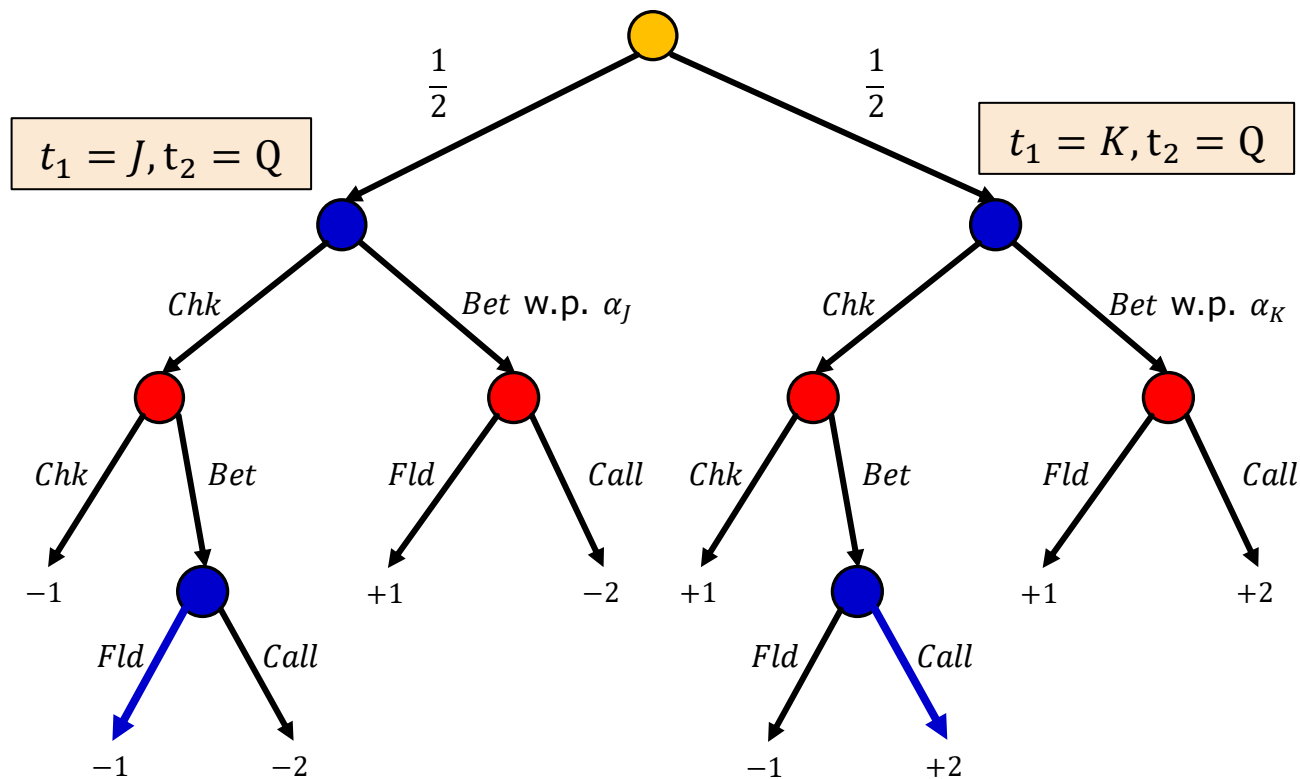
$$\frac{\alpha_J}{\alpha_J + \alpha_K} \cdot 2 + \frac{\alpha_K}{\alpha_J + \alpha_K} \cdot (-2) = 2 \cdot \frac{\alpha_J - \alpha_K}{\alpha_J + \alpha_K}$$



In order to play fold/call with positive probability, the utility from both must be the same. Thus:

$$-1 = 2 \cdot \frac{\alpha_J - \alpha_K}{\alpha_J + \alpha_K} \Rightarrow \alpha_K = 3\alpha_J$$

In equilibrium where P2 plays Fold/Call with positive probability, $\alpha_K = 3\alpha_J$.



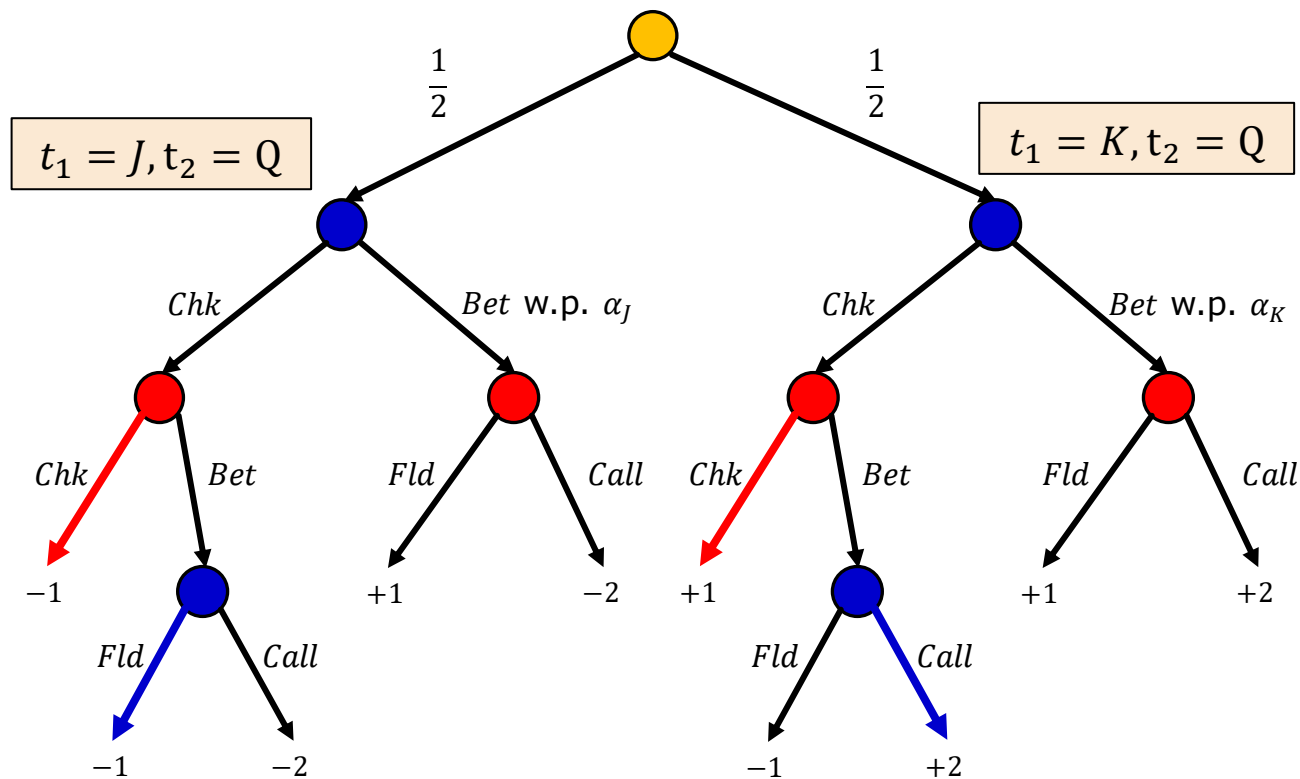
If $t_1 = J$, then $t_2 = Q, K$ with prob. $\frac{1}{2}$.

As previously argued, if $t_2 = K$ they will always call. If $t_2 = Q$, they call with probability δ_Q . Thus P1's expected utility from betting is:

$$\mathbb{E}[u_1(\text{bet with } J)] = \frac{1}{2} \cdot (-2) + \frac{1}{2} \left((1 - \delta_Q) \cdot 1 + \delta_Q \cdot (-2) \right) = -\frac{1}{2} - \frac{3}{2} \delta_Q$$

$$\mathbb{E}[u_1(\text{check with } J)] = -1$$

Equating the two we get $\frac{3}{2} \delta_Q = \frac{1}{2} \Rightarrow \delta_Q = \frac{1}{3}$.



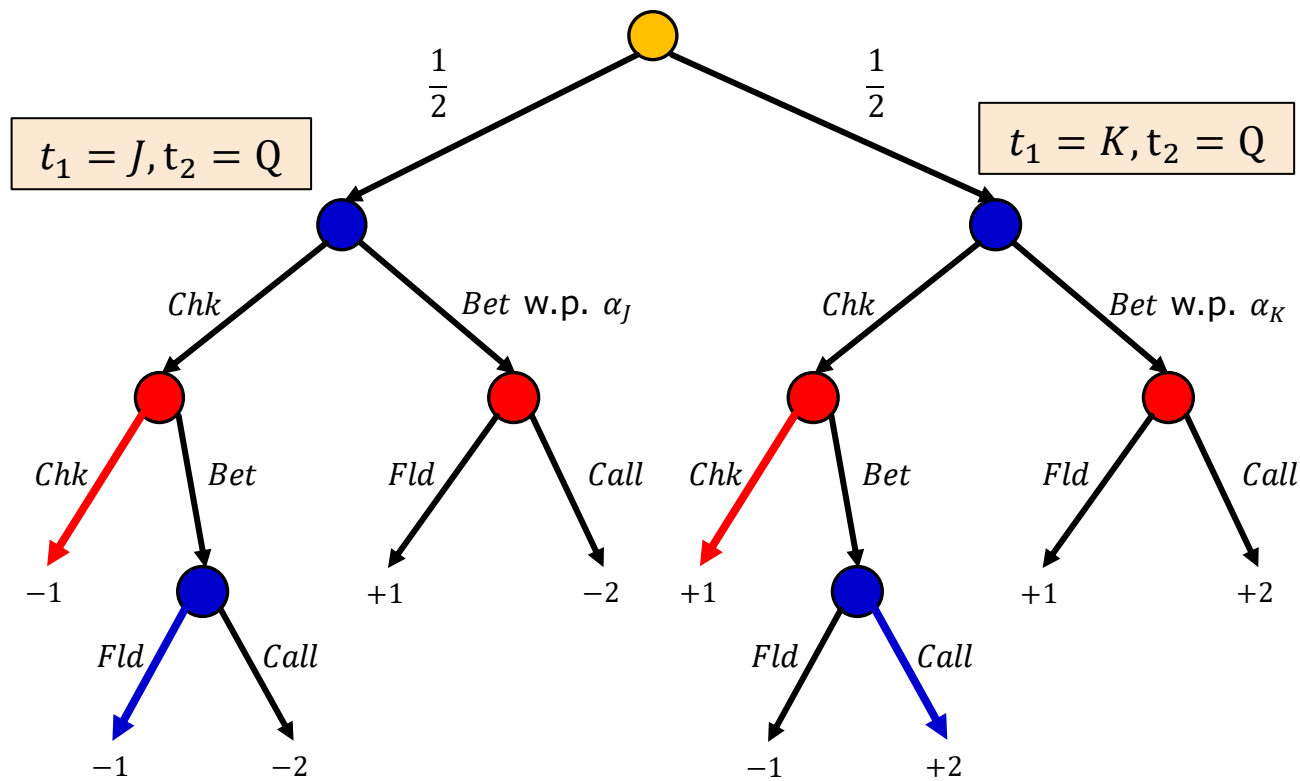
Let γ_Q be the probability that P2 bets when P1 checks.

$$\mathbb{E}[u_2(\text{Bet})] = \Pr[t_1 = J \mid \text{Chk}] \cdot 1 + \Pr[t_1 = K \mid \text{Chk}] \cdot (-2)$$

$$\mathbb{E}[u_2(\text{Chk})] = \Pr[t_1 = J \mid \text{Chk}] \cdot 1 + \Pr[t_1 = K \mid \text{Chk}] \cdot (-1)$$

Betting is weakly worse than checking.

Thus $\gamma_Q = 0$ in any Bayes-Nash equilibrium where $t_2 = Q$.



Suppose that $t_1 = K$. If they check, P2 will check as well and they get +1. If they bet, then P2 calls with probability $\delta_Q = \frac{1}{3}$ and their expected utility is strictly higher. Thus, when P1 has $t_1 = K$, they bet with probability $\alpha_K = 1$.

Thus, $\alpha_J = \frac{1}{3}$.